EMBEDDING DEPENDENCE OF LYUBEZNIK NUMBERS FOR NORMAL PROJECTIVE VARIETIES

SHENG TAN

Abstract. We construct an integral normal complex projective variety with two projective embeddings for which the Lyubeznik number $\lambda_{2,9}$ of the affine cone is 2 and 1, respectively. This answers a question of B. Wang [Proc. Amer. Math. Soc. 148 (2020), 2091–2096]. The main ingredient is a general construction that embeds any reduced projective scheme S into a normal projective variety X with a small resolution and transfers the relevant perverse cohomology together with the action of first Chern classes from S to X. The result follows by applying this construction to an example of Reichelt–Saito–Walther.

1. Introduction

Lyubeznik numbers are numerical invariants of local rings defined by Bass numbers of local cohomology modules. In characteristic zero, Lyubeznik proved that these numbers are finite and independent of the chosen regular local presentation [Lyu93]. Despite their algebraic definition, they often reflect the topology of the singularity. For complex isolated singularities, García López and Sabbah expressed them in terms of the Betti numbers of the link [GLS98]. Walther gave a topological interpretation in dimension two [Wal01], independently obtained by Kawasaki [Kaw00]. We refer to [NBWZ16] for a survey.

Given a projective scheme Z and an embedding $\iota : Z \hookrightarrow \mathbb{P}^N$, one can consider the Lyubeznik numbers of the local ring at the vertex of its affine cone. Since this local ring can depend on the embedding, Lyubeznik asked whether its Lyubeznik numbers depend only on Z [Lyu02, p. 133].

The answer depends on the characteristic. Zhang proved embedding independence in positive characteristic [Zha11]. Over $\mathbb{C}$, García López–Sabbah and Switala proved it for smooth projective varieties [GLS98, Swi15]. Reichelt, Saito, and Walther later expressed the Lyubeznik numbers in rows $k \geq 2$ in terms of maps induced by cup product with first Chern classes on the perverse cohomology of the dual constant complex [RSW21]. This description allowed them to construct reducible projective schemes with embedding-dependent Lyubeznik numbers.

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Wang then constructed irreducible projective varieties with embedding-dependent Lyubeznik numbers [Wan20]. His examples are not normal, and he asked whether the same phenomenon can occur for a normal projective variety [Wan20, Question 11]. For surfaces, normality implies embedding independence by [RWZ22, Theorem 3.20].

We answer Wang's question affirmatively. For an embedding $\iota : Z \hookrightarrow \mathbb{P}^N$, let A_ι be the local ring at the vertex of the affine cone defined by the homogeneous ideal of this embedding.

Theorem A. *There exist an integral normal complex projective variety X and two projective embeddings ι_1, ι_2 of X such that*

$$\lambda_{2,9}(A_{\iota_1}) = 2, \quad \lambda_{2,9}(A_{\iota_2}) = 1.$$

The variety X can be chosen rational and admits a projective small resolution from a smooth projective variety.

The two numerical values already occur in the reducible example of [RSW21, Section 2.2, Remark (i)], in the entry $\lambda_{2,6}$. Our construction places that example inside a different, normal variety. A small resolution identifies the relevant perverse cohomology sheaves, while the choice of ample line bundles preserves the ranks of the two maps induced by first Chern classes.

We prove this construction for an arbitrary reduced projective scheme. For such a scheme Z, put

$$\mathcal{P}_q(Z) = {}^p\mathcal{H}^{-q}(\mathbb{D}\mathbb{Q}_Z), \quad V_q^a(Z) = \mathbb{H}^a(Z, \mathcal{P}_q(Z)).$$

Here $\mathbb{D}$ is Verdier duality, and ${}^p\mathcal{H}$ denotes cohomology for the middle perverse t-structure. All sheaves have rational coefficients.

Theorem B. *Let S be a nonempty reduced complex projective scheme, and let L_1, L_2 be ample line bundles on S. There exist an integral normal projective variety X, a closed immersion $i : S \hookrightarrow X$, a positive integer a, and very ample line bundles T_1, T_2 on X such that*

$$i^*T_1 \simeq L_1^a, \quad i^*T_2 \simeq L_2^a.$$

The variety X is rational and has dimension $n > \dim S + 3$. It admits a projective small resolution $f : M \to X$ such that f is an isomorphism over $X \setminus i(S)$ and

$$\left(f^{-1}(i(S))\right)_{\mathrm{red}} \simeq S \times \mathbb{P}^1$$

over S. For every integer q with $q + 3 < n$, there is an isomorphism

$$\mathcal{P}_{q+3}(X) \simeq i_*\mathcal{P}_q(S)$$

compatible with the action of a cohomology class on X and its restriction to S.

The proof starts with an embedding $S \hookrightarrow B$ into a smooth projective variety. We construct a smooth variety M whose projection contracts exactly $S \times \mathbb{P}^1$, and let X be the normalization of its image. The scheme S embeds into X, and the fibers are unchanged after normalization. Since the exceptional

fibers are projective lines, the resulting distinguished triangle for $Rf_*\mathbb{Q}_M$, together with the smallness of $f : M \to X$, transfers the relevant perverse cohomology from S to X with a shift of three. The construction is compatible with the action of first Chern classes.

To extend the two polarizations, we embed S into a product of projective spaces using suitable powers of $L_1^N \otimes L_2^{-1}$ and $L_2^N \otimes L_1^{-1}$. Suitable ample line bundles on the product restrict to common powers of L_1 and L_2, and twisting their pullbacks by the relative $\mathcal{O}(1)$ of the construction gives the required line bundles on X.

Section 2 recalls the formula of Reichelt–Saito–Walther and gives the projective small resolution. Section 3 proves Theorem B and the resulting transfer of Lyubeznik numbers. Finally, Section 4 computes the example of Reichelt–Saito–Walther and applies the construction to prove Theorem A.

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2. Lyubeznik numbers and a projective small resolution

All schemes are of finite type over $\mathbb{C}$. We use the analytic topology for constructible sheaves. Let $D_c^b(Z, \mathbb{Q})$ denote the bounded derived category of $\mathbb{Q}$-constructible sheaves on Z. This category has the middle perverse t-structure, with cohomology functors ${}^p\mathcal{H}^a$. For an integral variety Z of dimension d, the intersection complex $\mathrm{IC}_Z \mathbb{Q}$ is normalized to restrict to $\mathbb{Q}[d]$ on the smooth locus. We use the basic properties of perverse sheaves from [BBD82]. In particular, the direct image under a closed immersion is perverse exact, and for a smooth morphism of relative dimension e, the functor $f^*[e]$ is perverse exact.

Definition 2.1. Let A be a quotient of a regular local $\mathbb{C}$-algebra, with residue field $\mathbb{C}$. Choose a surjection $R \twoheadrightarrow A$ from a regular local $\mathbb{C}$-algebra R of dimension m, and let I be its kernel. The Lyubeznik numbers of A are

$$\lambda_{k,j}(A) = \dim_{\mathbb{C}} \operatorname{Ext}_R^k(\mathbb{C}, H_I^{m-j}(R)),$$

where $H_I^{m-j}(R)$ denotes local cohomology with support in I.

These dimensions are finite and independent of the presentation [Lyu93]. In this paper, A is the vertex local ring of an affine cone. More precisely, for a closed immersion $\iota : Z \hookrightarrow \mathbb{P}^N$, let $I_\iota \subset \mathbb{C}[x_0, \ldots, x_N]$ be its homogeneous ideal and put

$$C_\iota(Z) = \operatorname{Spec}(\mathbb{C}[x_0, \ldots, x_N]/I_\iota), \quad A_\iota = \mathcal{O}_{C_\iota(Z),v},$$

where v is the vertex. Thus the local cohomological degree in Definition 2.1 is $N + 1 - j$. If L is very ample and ι is the embedding defined by its complete linear system, we also write A_L for A_ι.

For any reduced projective scheme Z, we use the notation

$$\mathcal{P}_q(Z) = {}^p\mathcal{H}^{-q}(\mathbb{D}\mathbb{Q}_Z), \quad V_q^a(Z) = \mathbb{H}^a(Z, \mathcal{P}_q(Z)).$$

The groups $V_q^a(Z)$ are finite-dimensional rational vector spaces. Cup product with the first Chern class of a line bundle L defines a map

$$c_1(L) : V_q^a(Z) \longrightarrow V_q^{a+2}(Z).$$

The following is [RSW21, Propositions 1–3], with the indices written in our notation.

Proposition 2.2 (Reichelt–Saito–Walther)**.** *Let $\iota : Z \hookrightarrow \mathbb{P}^N$ be a projective embedding of a reduced scheme, and set $L = \iota^*\mathcal{O}_{\mathbb{P}^N}(1)$. For $k \geq 2$ and $j \geq 1$, one has*

$$\begin{aligned} \lambda_{k,j}(A_\iota) = {} & \dim_{\mathbb{Q}} \operatorname{coker}\bigl(c_1(L) : V_{j-1}^{k-2}(Z) \longrightarrow V_{j-1}^{k}(Z)\bigr) \\ & + \dim_{\mathbb{Q}} \ker\bigl(c_1(L) : V_{j-1}^{k-1}(Z) \longrightarrow V_{j-1}^{k+1}(Z)\bigr). \end{aligned} \tag{2.1}$$

We will also use the action of cohomology on a constructible complex. A class $\alpha \in H^2(Z, \mathbb{Q})$ is a morphism $\mathbb{Q}_Z \to \mathbb{Q}_Z[2]$. Tensoring with a complex K defines its action on K and on $\mathbb{H}^\bullet(Z, K)$. This construction is natural in K. For a closed immersion $i : S \hookrightarrow Z$, its action on i_*K agrees, by the projection formula, with the action of $i^*\alpha$ on K.

We now turn to the geometric construction. Throughout the paper, if F is a vector bundle on a scheme B, $\mathbb{P}_B(F)$ denotes the projective bundle of one-dimensional subspaces of the fibers of F, with projection $\pi : \mathbb{P}_B(F) \to B$ and tautological line subbundle $\mathcal{O}_{\mathbb{P}_B(F)}(-1) \subset \pi^*F$.

Recall that a proper birational morphism $f : M \to X$ is small if

$$\operatorname{codim}_X \{x \in X : \dim f^{-1}(x) \geq e\} > 2e$$

for every positive integer e. If f is a small resolution, then M is smooth. For such a resolution, $Rf_*\mathbb{Q}_M[\dim M] \simeq \mathrm{IC}_X \mathbb{Q}$; see [dCM09, Remark 4.2.4]. We will also check this identification directly for the following construction.

Proposition 2.3. *Let B be a smooth integral projective variety, let E be a vector bundle of rank $r \geq 2$, and let $s \in H^0(B, E)$ be a section whose zero scheme S is reduced. There exist an integral normal projective variety X, a projective small resolution $f : M \to X$, a closed immersion $i : S \hookrightarrow X$, and a morphism $g : X \to B$ with the following properties. The composite $g \circ i$ is the original inclusion. The map f is an isomorphism over $X \setminus i(S)$, and*

$$\bigl(f^{-1}(i(S))\bigr)_{\mathrm{red}} \simeq S \times \mathbb{P}^1$$

over S. Moreover, $\dim X = \dim B + r + 1$. There is a g-ample line bundle R on X with $i^*R \simeq \mathcal{O}_S$. If B is rational, then M and X are rational.

Proof. Set $F = \mathcal{O}_B \oplus E \oplus E \oplus E$. Write the fiber coordinates on $\mathbb{P}_B(F)$ as $[t : x : y : z]$, where x, y, z have values in the corresponding copies of E. Let $[u : v]$ be the coordinates on $\mathbb{P}^1$. Define

$$M = \{xu = (z - ts)v,\ (z + ts)u = yv\} \subset \mathbb{P}_B(F) \times \mathbb{P}^1. \tag{2.2}$$

On $B \times \mathbb{P}^1$, the equations come from the map of vector bundles

$$\Phi : \mathrm{pr}_B^* F \longrightarrow \left(\mathrm{pr}_B^* E \otimes \mathrm{pr}_{\mathbb{P}^1}^* \mathcal{O}_{\mathbb{P}^1}(1)\right)^{\oplus 2},$$
$$(t, x, y, z) \longmapsto (ux - vz + vts,\ uz - vy + uts).$$

The map Φ is everywhere surjective. Indeed, on the chart $u \neq 0$, the variables z and x can be chosen successively to give any prescribed pair in the target. On $v \neq 0$, the variables z and y do the same. Equivalently, the coefficient matrix in x, y, z on each summand of E is

$$\begin{pmatrix} u & 0 & -v \\ 0 & -v & u \end{pmatrix},$$

whose maximal minors are $-uv, u^2, -v^2$. They do not vanish simultaneously.

Thus $K = \ker \Phi$ is a vector bundle of rank $r+1$, and $M = \mathbb{P}_{B\times\mathbb{P}^1}(K)$. In particular, M is smooth, integral, and projective, of dimension $\dim B + r + 1$.

Let Y_0 be the reduced image of the projection $f_0 : M \to \mathbb{P}_B(F)$. Since M is integral, so is Y_0. After trivializing E, the fiber over a point of Y_0 is the projectivization of the kernel of

$$\begin{pmatrix} x_1 & ts_1 - z_1 \\ z_1 + ts_1 & -y_1 \\ \vdots & \vdots \\ x_r & ts_r - z_r \\ z_r + ts_r & -y_r \end{pmatrix}. \tag{2.3}$$

The matrix has rank at most one on Y_0. Over the rank-one locus, a nonzero row determines the kernel line regularly, and the resulting local inverse maps agree on overlaps. Hence f_0 is an isomorphism there.

The matrix has rank zero precisely when $x = y = z = 0$ and $ts = 0$. Since the homogeneous coordinates cannot all vanish, $t \neq 0$ at such a point. The rank-zero locus is therefore

$$S_0 = S \times \{[1 : 0 : 0 : 0]\} \subset \mathbb{P}_B(F).$$

Its inverse image is $S \times \mathbb{P}^1$, and f_0 restricts there to the projection. All other fibers are single points. In particular, every fiber is connected, and the locus $t = 0$ is contained in the isomorphism locus. The latter is nonempty, so f_0 is birational.

Let $\nu : X \to Y_0$ be the normalization. It is finite, since Y_0 is of finite type over $\mathbb{C}$. The normality of M and the dominance of f_0 give a factorization

$$M \xrightarrow{f} X \xrightarrow{\nu} Y_0$$

by the universal property of normalization; see [Sta26, Section 29.55]. The variety X is integral, normal, and projective. The morphism f is projective, since its graph is closed in $M \times X$. Its image is closed and contains a dense open subset of X, so f is surjective.

For each complex point $y \in Y_0$, the connected fiber $f_0^{-1}(y)$ maps onto the finite set $\nu^{-1}(y)$. Hence $\nu^{-1}(y)$ consists of a single point. It follows that ν

is bijective on complex points and that f has the same underlying fibers as f_0. Moreover, $Y_0 \setminus S_0$ is isomorphic to a smooth open subset of M, so the normalization is an isomorphism over $Y_0 \setminus S_0$.

We next show that S embeds as a closed subscheme of X. The point $[u : v] = [0 : 1]$ of $\mathbb{P}^1$ defines a section

$$S \longrightarrow S \times \mathbb{P}^1 \subset M.$$

Composing this section with f gives a map $i : S \to X$, and $\nu \circ i$ is the original closed immersion $S \simeq S_0 \hookrightarrow Y_0$. The map i is a closed immersion. Indeed, it factors as a section of the separated morphism $X \times_{Y_0} S \to S$, followed by the closed immersion $X \times_{Y_0} S \hookrightarrow X$.

The reduced inverse image of $i(S)$ is $S \times \mathbb{P}^1$. On this reduced scheme, the morphisms f and $i \circ \mathrm{pr}_S$ agree on closed points, since ν is bijective on complex points. They therefore agree as morphisms. This proves the required description over S.

Let $g : X \to B$ be the structural map, and let

$$R = \nu^*\big(\mathcal{O}_{\mathbb{P}_B(F)}(1)|_{Y_0}\big).$$

The bundle R is g-ample, since finite pullback preserves relative ampleness. The restriction of $\mathcal{O}_{\mathbb{P}_B(F)}(1)$ to the section defined by the line subbundle $\mathcal{O}_B \subset F$ is trivial. Hence $i^*R \simeq \mathcal{O}_S$.

The only positive-dimensional fibers of f have dimension one and occur over $i(S)$. The locus in X over which these fibers occur has codimension

$$\dim X - \dim S = \dim B + r + 1 - \dim S \geq r + 1 > 2.$$

There are no fibers of dimension at least two. Thus f is small.

Finally, a projective bundle is birational to the product of its base with a projective space. Hence M is birational to $B \times \mathbb{P}^1 \times \mathbb{P}^r$. If B is rational, then M is rational, and so is the birational variety X. □

Remark 2.4. Every reduced closed subscheme S of a smooth projective variety B fits into the construction of Proposition 2.3. Choose a line bundle H such that $\mathcal{I}_S \otimes H$ is globally generated. A finite set of generators defines a section of $H^{\oplus r}$ whose zero scheme is S. Adding zero components to this section increases r without changing its zero scheme.

3. Perverse cohomology and polarizations

Use the notation of Proposition 2.3, and set $n = \dim X$.

Lemma 3.1. *There is a distinguished triangle in $D^b_c(X, \mathbb{Q})$,*

$$\mathbb{Q}_X \longrightarrow Rf_*\mathbb{Q}_M \longrightarrow i_*\mathbb{Q}_S[-2] \xrightarrow{+1}, \tag{3.1}$$

where the first map is the canonical adjunction morphism $\mathbb{Q}_X \to Rf_*\mathbb{Q}_M$.

Proof. Proper base change and the description of the fibers give

$$R^0 f_*\mathbb{Q}_M \simeq \mathbb{Q}_X, \quad R^1 f_*\mathbb{Q}_M = 0, \quad R^a f_*\mathbb{Q}_M = 0 \quad (a > 2).$$

Since every fiber is connected, the first isomorphism is induced by the canonical adjunction morphism $\mathbb{Q}_X \to Rf_*\mathbb{Q}_M$. The sheaf $R^2f_*\mathbb{Q}_M$ is supported on $i(S)$. Over $i(S)$, the reduced inverse image is $S \times \mathbb{P}^1$, so proper base change identifies the restriction of $R^2f_*\mathbb{Q}_M$ to S with $\mathbb{Q}_S$. These local identifications agree globally: the class pulled back from $\mathcal{O}_{\mathbb{P}^1}(1)$ generates the second cohomology of every exceptional fiber. Hence

$$R^2f_*\mathbb{Q}_M \simeq i_*\mathbb{Q}_S.$$

The cone of the canonical morphism $\mathbb{Q}_X \to Rf_*\mathbb{Q}_M$ has exactly one nonzero ordinary cohomology sheaf, namely $i_*\mathbb{Q}_S$ in degree two. It is therefore isomorphic to $i_*\mathbb{Q}_S[-2]$, which gives (3.1). □

Lemma 3.2. *There is an isomorphism*

$$Rf_*\mathbb{Q}_M[n] \simeq \mathrm{IC}_X\,\mathbb{Q}.$$

Consequently, $Rf_*\mathbb{D}\mathbb{Q}_M$ has perverse cohomology only in degree $-n$.

Proof. Put $K = Rf_*\mathbb{Q}_M[n]$. By Lemma 3.1, the only nonzero ordinary cohomology sheaves of K are

$$\mathcal{H}^{-n}(K) = \mathbb{Q}_X, \quad \mathcal{H}^{2-n}(K) = i_*\mathbb{Q}_S.$$

Since $\dim S \leq n-3$, these sheaves satisfy the support inequalities for the perverse t-structure. Hence $K \in {}^pD^{\leq 0}(X)$. Since M is smooth, proper duality gives $\mathbb{D}K \simeq K$. Applying the same support inequalities to the dual gives $K \in {}^pD^{\geq 0}(X)$. Thus K is perverse.

Proper base change and the identification $(f^{-1}(i(S)))_{\mathrm{red}} \simeq S \times \mathbb{P}^1$ give

$$i^*K \simeq \mathbb{Q}_S[n] \oplus \mathbb{Q}_S[n-2].$$

Since $\dim S \leq n-3$, we have $i^*K \in {}^pD^{\leq -1}(S)$. By duality, $i^!K \in {}^pD^{\geq 1}(S)$. Hence K has neither a perverse quotient nor a perverse subobject supported on $i(S)$. On $X \setminus i(S)$, the morphism f is an isomorphism, so K restricts to the constant perverse sheaf $\mathbb{Q}_{X\setminus i(S)}[n]$. The characterization of the intermediate extension therefore gives $K \simeq \mathrm{IC}_X\,\mathbb{Q}$.

Finally, $\mathbb{D}\mathbb{Q}_M \simeq \mathbb{Q}_M[2n]$. Hence $Rf_*\mathbb{D}\mathbb{Q}_M \simeq \mathrm{IC}_X\,\mathbb{Q}[n]$, which has perverse cohomology only in degree $-n$. □

Proposition 3.3. *For every integer q with $q+3 < n$, there is an isomorphism*

$$\mathcal{P}_{q+3}(X) \simeq i_*\mathcal{P}_q(S). \tag{3.2}$$

Under the induced isomorphism $V^\bullet_{q+3}(X) \simeq V^\bullet_q(S)$, cup product by $\alpha \in H^2(X,\mathbb{Q})$ corresponds to cup product by $i^*\alpha$.

Proof. Apply Verdier duality to (3.1). Since f and i are proper, the resulting triangle is

$$i_*\mathbb{D}\mathbb{Q}_S[2] \longrightarrow Rf_*\mathbb{D}\mathbb{Q}_M \longrightarrow \mathbb{D}\mathbb{Q}_X \longrightarrow i_*\mathbb{D}\mathbb{Q}_S[3].$$

By Lemma 3.2, the middle term has perverse cohomology only in degree $-n$. For $a > -n$, the connecting morphism therefore induces an isomorphism

$${}^p\mathcal{H}^a(\mathbb{D}\mathbb{Q}_X) \simeq i_*\,{}^p\mathcal{H}^{a+3}(\mathbb{D}\mathbb{Q}_S).$$

Taking $a = -(q+3)$ proves (3.2).

This is an isomorphism of perverse sheaves on X. The action of $H^2(X, \mathbb{Q})$ on hypercohomology is natural with respect to such morphisms. On a complex supported on $i(S)$, the projection formula identifies it with the action of the restricted class. This proves the compatibility. □

Lemma 3.4. *Let B be a projective scheme, let F be a vector bundle, let $\pi : \mathbb{P}_B(F) \to B$ be the projection, and let H be a line bundle on B. If $F^\vee \otimes H$ is a direct sum of very ample line bundles, then*

$$\mathcal{O}_{\mathbb{P}_B(F)}(1) \otimes \pi^* H$$

is very ample.

Proof. Write $F^\vee \otimes H = \bigoplus_{a=1}^m H_a$, where each H_a is very ample. A line $\ell \subset F_b$ determines a one-dimensional quotient

$$F_b^\vee \otimes H_b \longrightarrow \ell^\vee \otimes H_b.$$

Thus $\mathbb{P}_B(F)$ is naturally identified with the projective bundle of one-dimensional quotients of $F^\vee \otimes H$, and under this identification $\mathcal{O}_{\mathbb{P}_B(F)}(1) \otimes \pi^* H$ is the tautological quotient line bundle. Its sections are therefore induced by the sections of the H_a.

These sections generate the line bundle and separate points in each projective fiber. If two points lie over different points of B, choose a summand whose quotient coordinate is nonzero at the first point. Since this summand is very ample, there is a section that vanishes at the second base point but not at the first, and the induced section separates the two points.

The sections also separate tangent vectors. Tangent vectors along a fiber are separated by the complete linear system of $\mathcal{O}(1)$ on that fiber. For a tangent vector whose image in B is nonzero, choose a summand whose quotient coordinate at the point is nonzero. Since this summand is very ample, there is a section that vanishes at the base point and has nonzero derivative along the image of the tangent vector. The induced section then separates the tangent vector. Hence $\mathcal{O}_{\mathbb{P}_B(F)}(1) \otimes \pi^* H$ is very ample. □

Proof of Theorem B. Choose an integer $N > 1$ such that $L_1^N \otimes L_2^{-1}$ and $L_2^N \otimes L_1^{-1}$ are ample. Choose a common positive integer p such that

$$C_1 = (L_1^N \otimes L_2^{-1})^p, \quad C_2 = (L_2^N \otimes L_1^{-1})^p$$

are very ample. Their complete linear systems define a closed immersion

$$S \hookrightarrow B = \mathbb{P}^{N_1} \times \mathbb{P}^{N_2}.$$

On B, put $H_1 = \mathcal{O}_B(N, 1)$ and $H_2 = \mathcal{O}_B(1, N)$. Their restrictions to S are

$$H_1|_S \simeq L_1^{p(N^2-1)}, \quad H_2|_S \simeq L_2^{p(N^2-1)}.$$

Choose a positive integer d such that $\mathcal{I}_S(d, d)$ is globally generated. A finite set of generators defines a section s of $E = \mathcal{O}_B(d, d)^{\oplus r}$ whose zero scheme is S. Adding zero components if necessary, we may assume that

$r \geq 4$. Apply Proposition 2.3 to obtain $f : M \to X$, $i : S \hookrightarrow X$, $g : X \to B$, and R. Since B is rational, so is X. Moreover,

$$\dim X = \dim B + r + 1 > \dim S + 3.$$

The assertion about perverse cohomology and its compatibility with the action of first Chern classes follow from Proposition 3.3.

It remains to construct the required line bundles on X. Define

$$\widetilde{T}_1 = R \otimes g^* H_1^{d+1}, \quad \widetilde{T}_2 = R \otimes g^* H_2^{d+1}.$$

The line bundle $\widetilde{T}_1$ is the pullback from Y_0 of the restriction of

$$\mathcal{O}_{\mathbb{P}_B(F)}(1) \otimes \pi^* \mathcal{O}_B(N(d+1), d+1), \quad F = \mathcal{O}_B \oplus E^{\oplus 3}.$$

We have

$$F^\vee \otimes \mathcal{O}_B(N(d+1), d+1) \simeq \mathcal{O}_B(N(d+1), d+1) \oplus \mathcal{O}_B(N(d+1) - d, 1)^{\oplus 3r}.$$

Both types of summands are very ample. Hence Lemma 3.4 shows that

$$\mathcal{O}_{\mathbb{P}_B(F)}(1) \otimes \pi^* \mathcal{O}_B(N(d+1), d+1)$$

is very ample. Its restriction to Y_0 is therefore very ample, and its pullback under the finite normalization $\nu : X \to Y_0$ is ample [Sta26, Lemma 30.17.2]. Thus $\widetilde{T}_1$ is ample. The same argument, with the two factors exchanged, shows that $\widetilde{T}_2$ is ample.

Choose a common positive integer t such that $T_1 = \widetilde{T}_1^t$ and $T_2 = \widetilde{T}_2^t$ are very ample. Since $i^* R \simeq \mathcal{O}_S$, we have

$$i^* T_\nu \simeq L_\nu^a, \quad a = t(d+1)p(N^2 - 1), \quad \nu = 1, 2.$$

This proves the theorem. □

Corollary 3.5. *Let S be a nonempty reduced complex projective scheme, and let L_1, L_2 be very ample line bundles on S. Choose X, T_1, T_2 as in Theorem B. Then*

$$\lambda_{k,j+3}(A_{T_\nu}) = \lambda_{k,j}(A_{L_\nu})$$

for $\nu = 1, 2$, $k \geq 2$, and $1 \leq j \leq \dim S + 1$.

Proof. Set $q = j - 1$. Since $q \leq \dim S$, we have $q + 3 < \dim X$. By Theorem B,

$$\mathcal{P}_{j+2}(X) \simeq i_* \mathcal{P}_{j-1}(S).$$

Under the induced identifications of hypercohomology groups, the action of $c_1(T_\nu)$ corresponds to the action of

$$c_1(L_\nu^a) = a\, c_1(L_\nu).$$

Since a is nonzero in $\mathbb{Q}$, multiplication by a does not change the kernels or cokernels of these maps. Applying (2.1) to X and S gives the desired equality. □

4. Proof of the main theorem

We now prove Theorem A. We begin with the example of [RSW21, Sections 2.1–2.2], with the dimensions chosen in Remark (i) following Section 2.2. We describe the corresponding reduced projective scheme and compute the relevant maps induced by first Chern classes in fixed bases.

Let $Z = \mathbb{P}^1 \times \mathbb{P}^1$, let $D \subset Z$ be the diagonal, and choose a section σ of $\mathcal{O}_Z(1,1)$ whose zero divisor is D. Inside $\operatorname{Tot} \mathcal{O}_Z(1,1)$, let Z_0 be the zero section and let Z_1 be the graph of σ. Set

$$S_1 = Z_0 \cup Z_1, \quad p : S_1 \longrightarrow Z.$$

The projection p is finite. Locally, S_1 is defined by $w(w-\sigma) = 0$. Since D is smooth, the two smooth components meet transversely along D. Moreover, S_1 is projective, since it is a closed subscheme of a projective bundle compactifying $\operatorname{Tot} \mathcal{O}_Z(1,1)$.

Lemma 4.1. *The line bundle $p^*\mathcal{O}_Z(a,b)$ is very ample on S_1 whenever $a, b \geq 1$.*

Proof. The line bundle $\mathcal{O}_Z(a,b)$ is very ample, so the pullbacks of its sections separate points of S_1 with distinct images in Z and tangent vectors with nonzero image in Z. Let w denote the tautological fiber coordinate on the total space of $\mathcal{O}_Z(1,1)$. For any section h of $\mathcal{O}_Z(a-1,b-1)$, the product wh restricts to a section of $p^*\mathcal{O}_Z(a,b)$ on S_1.

Since $\mathcal{O}_Z(a-1,b-1)$ is globally generated, for any point of $Z \setminus D$ we may choose h that does not vanish there. The section wh vanishes at the point of Z_0 and is nonzero at the point of Z_1 lying over the same point of Z. Thus it separates these two points.

At a point of D, choose local coordinates in which σ is one of the parameters on Z. The tangent space of S_1, locally defined by $w(w-\sigma) = 0$, has one additional direction represented by w besides the two directions coming from Z. Choosing h nonzero at the point, the differential of wh is nonzero in this additional direction. Hence the sections of $p^*\mathcal{O}_Z(a,b)$ separate points and tangent vectors on S_1, so $p^*\mathcal{O}_Z(a,b)$ is very ample. □

Let h_1, h_2 be the hyperplane classes on Z, and use the same notation for their pullbacks to each component of S_1. Put

$$A_1 = p^*\mathcal{O}_Z(1,1), \quad A_2 = p^*\mathcal{O}_Z(2,1).$$

Lemma 4.2. *One has*

$$H_4(S_1,\mathbb{Q}) \simeq \mathbb{Q}^2, \quad H_2(S_1,\mathbb{Q}) \simeq \mathbb{Q}^3, \quad H_1(S_1,\mathbb{Q}) = H_3(S_1,\mathbb{Q}) = 0.$$

In fixed bases of $H_4(S_1,\mathbb{Q})$ and $H_2(S_1,\mathbb{Q})$, cap product by $c_1(p^*\mathcal{O}_Z(a,b))$ is the matrix

$$M(a,b) = \begin{pmatrix} a & b \\ b & b \\ 0 & a-b \end{pmatrix}. \tag{4.1}$$

Proof. Apply the Mayer–Vietoris sequence in homology to $S_1 = Z_0 \cup Z_1$, where $Z_0 \cap Z_1 = D \simeq \mathbb{P}^1$. We obtain

$$H_4(S_1, \mathbb{Q}) = H_4(Z_0, \mathbb{Q}) \oplus H_4(Z_1, \mathbb{Q})$$

and an exact sequence

$$0 \longrightarrow H_3(S_1, \mathbb{Q}) \longrightarrow H_2(D, \mathbb{Q}) \longrightarrow H_2(Z_0, \mathbb{Q}) \oplus H_2(Z_1, \mathbb{Q}) \longrightarrow H_2(S_1, \mathbb{Q}) \longrightarrow 0.$$

The map $H_2(D, \mathbb{Q}) \to H_2(Z_0, \mathbb{Q}) \oplus H_2(Z_1, \mathbb{Q})$ sends a generator to $([D], -[D])$ and is injective. Hence $H_3(S_1, \mathbb{Q}) = 0$ and

$$H_2(S_1, \mathbb{Q}) = \frac{H_2(Z_0, \mathbb{Q}) \oplus H_2(Z_1, \mathbb{Q})}{\mathbb{Q}([D], -[D])}.$$

Similarly, the map $H_0(D, \mathbb{Q}) \to H_0(Z_0, \mathbb{Q}) \oplus H_0(Z_1, \mathbb{Q})$ is injective. Since Z_0, Z_1, and D have no first homology, it follows that $H_1(S_1, \mathbb{Q}) = 0$.

Let $[Z_0], [Z_1]$ be a basis of $H_4(S_1, \mathbb{Q})$. On the two smooth components, define

$$e_i = h_i \cap [Z_0], \quad e_i' = h_i \cap [Z_1], \quad i = 1, 2.$$

Since the diagonal has class $h_1 + h_2$, the relation in the quotient is

$$e_1 + e_2 - e_1' - e_2' = 0.$$

We use the images of e_1, e_2, e_1' as a basis of $H_2(S_1, \mathbb{Q})$. Cap product by $ah_1 + bh_2$ sends $[Z_0]$ to $ae_1 + be_2$ and sends $[Z_1]$ to

$$ae_1' + be_2' = be_1 + be_2 + (a - b)e_1'.$$

These are precisely the two columns of (4.1). □

For the second factor, let $V = \mathrm{Bl}_P \mathbb{P}^5$ be the blowup of $\mathbb{P}^5$ at a point P, and let $\rho : V \to \mathbb{P}^4$ be the projection from P. The morphism ρ is a $\mathbb{P}^1$-bundle. Let Σ_0 be the exceptional section, and let Σ_∞ be the section defined by a hyperplane avoiding P. Choose a linear subspace $W \simeq \mathbb{P}^2 \subset \mathbb{P}^4$ and set

$$W_1 = \rho^{-1}(W), \quad W_2 = \Sigma_0 \sqcup \Sigma_\infty, \quad S_2 = W_1 \cup W_2.$$

The component W_1 has dimension three, while the two components of W_2 have dimension four. Let H denote the pullback of the hyperplane class on $\mathbb{P}^5$, and let E_P be the exceptional divisor. The line bundle

$$Q = \mathcal{O}_V(2H - E_P)|_{S_2}$$

is very ample. Indeed, V is a closed incidence variety in $\mathbb{P}^5 \times \mathbb{P}^4$, and the restriction of $\mathcal{O}(1, 1)$ to V is $\mathcal{O}_V(2H - E_P)$.

Lemma 4.3. *The only nonzero perverse cohomology sheaves $\mathcal{P}_q(S_2)$ are $\mathcal{P}_3(S_2)$ and $\mathcal{P}_4(S_2)$. Moreover,*

$$V_3^a(S_2) \simeq \begin{cases} \mathbb{Q}, & a = -3, 2, \\ 0, & \text{otherwise.} \end{cases}$$

Every class in $H^2(S_2, \mathbb{Q})$ acts trivially on $V_3^\bullet(S_2)$.

Proof. Set $U = W_1 \setminus W_2$, and let $j : U \hookrightarrow W_1$, $\ell : W_1 \hookrightarrow S_2$, and $\eta : W_2 \hookrightarrow S_2$ be the inclusions. The variety W_1 is the blowup of $\mathbb{P}^3$ at P. Removing the exceptional divisor and the hyperplane section at infinity gives

$$U \simeq \mathbb{A}^3 \setminus \{0\}.$$

The complement of U in the smooth threefold W_1 is a Cartier divisor, so j is an affine open immersion. Hence Rj_* is perverse exact [BBD82, Corollary 4.1.3].

The exact sequence of constant sheaves associated with $S_2 \setminus W_2 = U$ gives, after Verdier duality, a distinguished triangle

$$\eta_* \mathbb{DQ}_{W_2} \longrightarrow \mathbb{DQ}_{S_2} \longrightarrow \ell_* Rj_* \mathbb{DQ}_U \xrightarrow{+1} .$$

Since W_2 is smooth of pure dimension four, the first term has perverse cohomology only in degree -4. Similarly, $\mathbb{DQ}_U[-3] = \mathbb{Q}_U[3]$ is perverse, and since j is affine, the third term has perverse cohomology only in degree -3. The perverse long exact sequence therefore gives

$$\mathcal{P}_3(S_2) \simeq \ell_* Rj_* \mathbb{Q}_U[3], \quad \mathcal{P}_4(S_2) \simeq \eta_* \mathbb{Q}_{W_2}[4],$$

and all other $\mathcal{P}_q(S_2)$ vanish.

Taking hypercohomology of the first isomorphism gives

$$V_3^a(S_2) \simeq H^{a+3}(\mathbb{A}^3 \setminus \{0\}, \mathbb{Q}).$$

The punctured affine space retracts onto a five-sphere, so its cohomology is one-dimensional in degrees zero and five and vanishes in all other degrees. This proves the formula for $V_3^a(S_2)$. If $\alpha \in H^2(S_2, \mathbb{Q})$, cup product by α sends the two possibly nonzero groups $V_3^{-3}(S_2)$ and $V_3^2(S_2)$ to $V_3^{-1}(S_2)$ and $V_3^4(S_2)$, respectively. Both target groups vanish, so α acts trivially on $V_3^\bullet(S_2)$. □

Define

$$S = S_1 \times S_2, \quad L_1 = A_1 \boxtimes Q, \quad L_2 = A_2 \boxtimes Q. \tag{4.2}$$

The scheme S is reduced and projective, with irreducible components of dimensions five and six. Both L_1 and L_2 are very ample.

Proposition 4.4. *For the scheme S in (4.2),*

$$\dim V_5^0(S) = 2, \quad \dim V_5^2(S) = 3, \quad V_5^1(S) = V_5^3(S) = 0.$$

After fixing the identifications with the homology groups in Lemma 4.2, the map $V_5^0(S) \to V_5^2(S)$ induced by a line bundle $p^*\mathcal{O}_Z(a,b) \boxtimes Q'$ is given by $M(a,b)$, for any line bundle Q' on S_2. In particular,

$$\lambda_{2,6}(A_{L_1}) = 2, \quad \lambda_{2,6}(A_{L_2}) = 1.$$

Proof. The scheme S_1 is a local complete intersection of pure dimension two. Thus $\mathbb{Q}_{S_1}[2]$ is perverse, and

$$\mathcal{P}_2(S_1) = \mathbb{DQ}_{S_1}[-2], \quad \mathcal{P}_q(S_1) = 0 \quad (q \neq 2).$$

This is also the description used in [RSW21, Section 2.1]. External products over $\mathbb{Q}$ are perverse exact, and Verdier duality commutes with external products. Together with Lemma 4.3, this gives

$$\mathcal{P}_5(S) = \mathcal{P}_2(S_1) \boxtimes \mathcal{P}_3(S_2).$$

Since S_1 is compact,

$$\mathbb{H}^a(S_1, \mathcal{P}_2(S_1)) \simeq H_{2-a}(S_1, \mathbb{Q}).$$

By the Künneth formula, the two possible contributions to $V_5^r(S)$ come from degrees 2 and -3 on the second factor. They are, respectively,

$$H_{4-r}(S_1, \mathbb{Q}) \otimes V_3^2(S_2) \quad \text{and} \quad H_{-1-r}(S_1, \mathbb{Q}) \otimes V_3^{-3}(S_2).$$

For $0 \leq r \leq 3$, the second term vanishes. Fixing a nonzero generator of $V_3^2(S_2)$ therefore gives an identification

$$V_5^r(S) \simeq H_{4-r}(S_1, \mathbb{Q}) \quad (0 \leq r \leq 3).$$

The dimension statements now follow from Lemma 4.2.

Under these identifications, the first Chern class coming from S_1 acts by cap product on S_1, while the first Chern class coming from S_2 acts trivially by Lemma 4.3. Hence the map from $V_5^0(S)$ to $V_5^2(S)$ is exactly $M(a, b)$.

The three maximal minors of $M(a, b)$ are $b(a-b)$, $a(a-b)$, and $b(a-b)$. For positive a, b, the matrix has rank one when $a = b$ and rank two when $a \neq b$. In particular,

$$M(1,1) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad M(2,1) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$$

have ranks one and two, respectively. Since $\dim V_5^2(S) = 3$ and $V_5^1(S) = V_5^3(S) = 0$, (2.1) gives

$$\lambda_{2,6}(A_{L_\nu}) = \dim \operatorname{coker}\big(c_1(L_\nu) : V_5^0(S) \to V_5^2(S)\big).$$

The two cokernel dimensions are $3 - 1 = 2$ and $3 - 2 = 1$, respectively. □

We now apply the construction of Theorem B to this example and complete the proof of Theorem A.

Proof of Theorem A. Fix S, L_1, L_2 as in (4.2). The line bundles

$$C_1 = L_1^3 \otimes L_2^{-1} = p^*\mathcal{O}_Z(1,2) \boxtimes Q^2, \quad C_2 = L_2^3 \otimes L_1^{-1} = p^*\mathcal{O}_Z(5,2) \boxtimes Q^2$$

are very ample by Lemma 4.1. Their complete linear systems define a closed immersion

$$S \hookrightarrow B = \mathbb{P}^{N_1} \times \mathbb{P}^{N_2}, \quad N_\nu = h^0(S, C_\nu) - 1.$$

Fix bases of these spaces of sections. Choose a positive integer d such that $\mathcal{I}_S(d, d)$ is globally generated, and choose generators $s_1, \ldots, s_r$. Adding zero components if necessary, we may assume that $r \geq 4$. Set

$$E = \mathcal{O}_B(d, d)^{\oplus r}, \quad s = (s_1, \ldots, s_r), \quad F = \mathcal{O}_B \oplus E^{\oplus 3}.$$

The zero scheme of s is the given reduced scheme S.

Let $M \subset \mathbb{P}_B(F) \times \mathbb{P}^1$ be defined by (2.2), let Y_0 be its reduced image in $\mathbb{P}_B(F)$, and let X be the normalization of Y_0. By Proposition 2.3, X is integral, normal, and projective. It has a projective small resolution $f : M \to X$, a closed immersion $i : S \hookrightarrow X$, and a map $g : X \to B$. Since B is rational, both M and X are rational. Moreover,

$$n = \dim X = \dim B + r + 1 \geq 11.$$

In particular, $8 < n$.

Let R be the line bundle from Proposition 2.3, and put

$$\widetilde{T}_1 = R \otimes g^*\mathcal{O}_B(3(d+1), d+1), \quad \widetilde{T}_2 = R \otimes g^*\mathcal{O}_B(d+1, 3(d+1)).$$

These are the ample line bundles constructed in the proof of Theorem B, with $N = 3$ and $p = 1$. For $\widetilde{T}_1$, the relevant bundle on $\mathbb{P}_B(F)$ is

$$\mathcal{O}_{\mathbb{P}_B(F)}(1) \otimes \pi^*\mathcal{O}_B(3d+3, d+1).$$

The corresponding bundle

$$F^\vee \otimes \mathcal{O}_B(3d+3, d+1)$$

is a direct sum of $\mathcal{O}_B(3d+3, d+1)$ and copies of $\mathcal{O}_B(2d+3, 1)$. All these summands are very ample, so Lemma 3.4 applies. The restriction to Y_0 is therefore very ample, and its pullback to X under the finite normalization $\nu : X \to Y_0$ is ample. Thus $\widetilde{T}_1$ is ample. The same argument, with the two factors exchanged, shows that $\widetilde{T}_2$ is ample.

Choose a common positive integer t such that

$$T_1 = \widetilde{T}_1^t, \quad T_2 = \widetilde{T}_2^t$$

are very ample. Let ι_ν be the embedding of X defined by the complete linear system of T_ν. All the data defining X were fixed before this choice of embedding.

Since $i^*R \simeq \mathcal{O}_S$ and

$$C_1^3 \otimes C_2 = L_1^8, \quad C_1 \otimes C_2^3 = L_2^8,$$

we obtain

$$i^*T_1 \simeq L_1^{8(d+1)t}, \quad i^*T_2 \simeq L_2^{8(d+1)t}.$$

By Proposition 3.3, with $q = 5$,

$$\mathcal{P}_8(X) \simeq i_*\mathcal{P}_5(S).$$

Consequently,

$$\dim V_8^0(X) = 2, \quad \dim V_8^2(X) = 3, \quad V_8^1(X) = V_8^3(X) = 0.$$

By the compatibility with the action of first Chern classes in Proposition 3.3, the two maps from $V_8^0(X)$ to $V_8^2(X)$ correspond to

$$8(d+1)t\, M(1,1) \quad \text{and} \quad 8(d+1)t\, M(2,1),$$

which have ranks one and two, respectively.

Finally, by Proposition 2.2 one has

$$\begin{aligned}\lambda_{2,9}(A_{\iota_\nu}) = &\dim \operatorname{coker}\big(c_1(T_\nu) : V_8^0(X) \longrightarrow V_8^2(X)\big)\\ &+ \dim \ker\big(c_1(T_\nu) : V_8^1(X) \longrightarrow V_8^3(X)\big).\end{aligned}$$

The second summand is zero for both embeddings, while the first is 2 for ι_1 and 1 for ι_2. This proves the theorem. □

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School of Mathematical Sciences, Capital Normal University, Beijing 100048, China

Email address: `tansheng2018@outlook.com`